Scholarship Management System

(Case Study 61 | C++ Final Project)

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Problem Statement

Problem

Scholarship processing involves manual eligibility checks and fund tracking, causing delays and errors.

Objectives

Streamline scholarship allocation and reduce processing errors using an automated system.

Functional Requirements

Student eligibility verification, scholarship allocation, and real-time status tracking.

Deliverables

Working C++ scholarship system with full reports and documentation.

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Case Study 61.Scholarship Management System

🧩 **Problem Statement**

Scholarship processing involves eligibility verification and fund tracking. This system automates scholarship management.

🎯 **Objectives**

- To streamline scholarship allocation
- To reduce processing errors

⚙️ **Functional Requirements**

- Student eligibility verification
- Scholarship allocation
- Status tracking

📦 **Deliverables**

- Scholarship system
- Reports and documentation

Our Solution

Scholarship Class

- name, minGPA, maxIncome, fundAmount
- checkEligibility(gpa, income)
- Returns status string

Student Class

- studentID, name, gpa, familyIncome
- scholarshipStatus, allocatedFunds
- displayStatus() method

ScholarshipSystem

- addStudent() with duplicate ID check
- processScholarships() – auto-match
- displayReport() + trackStudentByID()

Built-in Scholarships:

Merit Base — GPA ≥ 8.5, Income ≤ ₹8,00,000 → Fund: ₹50,000

Need Base — GPA ≥ 7.0, Income ≤ ₹4,00,000 → Fund: ₹30,000

Working / Method

STEP 1

Add Student

User enters ID, name, GPA & income.
System validates input & checks for duplicate ID.

STEP 2

Process Scholarships

System compares each student's GPA & income
against all defined scholarship criteria.

STEP 3

Eligibility Check

checkEligibility() returns ELIGIBLE, 'GPA too low', 'Income exceeds limit', or both.

STEP 4

Allocation & Report

Approved students get funds assigned.
Full report and individual tracking available.

Conclusion

Objective Achieved

- Fully automated scholarship
- eligibility check
- fund allocation
- eliminating manual errors.

OOP Principles Applied

Three well-structured classes —
- Scholarship,
- Student,
- ScholarshipSystem — with encapsulation.

Robust Validation

Input guards prevent
- duplicate IDs,
- invalid GPA
- Invalid income values

Future Scope

- File storage
- dynamic scholarship addition
- GUI interface
- advanced ranking algorithms.

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Thank You

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